

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

03

10 questions · ~15 min

Q1

GEOMETRY AND TRIGONOMETRY

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is 3 times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

- A. 10°
- B. 30°
- C. 40°
- D. 90°

Q2

GEOMETRY AND TRIGONOMETRY

The length of each side of a square is 94 centimeters (cm). Which expression gives the area, in cm^2 , of the square?

- A. $2 \cdot 94$
- B. $2 \cdot 94 \cdot 94$
- C. $4 \cdot 94$
- D. $94 \cdot 94$

Q3

GEOMETRY AND TRIGONOMETRY

A circle in the xy -plane has the equation $(x - 13)^2 + (y - k)^2 = 64$. Which of the following gives the center of the circle and its radius?

- A. The center is at $(13, k)$ and the radius is 8.
- B. The center is at $(k, 13)$ and the radius is 8.
- C. The center is at $(k, 13)$ and the radius is 64.
- D. The center is at $(13, k)$ and the radius is 64.

Q4

GEOMETRY AND TRIGONOMETRY

Triangle FGH is similar to triangle JKL , where angle F corresponds to angle J and angles G and K are right angles. If $\sin F = \frac{308}{317}$, what is the value of $\sin(J)$?

- A. $\frac{75}{317}$
- B. $\frac{308}{317}$
- C. $\frac{317}{308}$
- D. $\frac{317}{75}$

GRID-IN GEOMETRY AND TRIGONOMETRY

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

GEOMETRY AND TRIGONOMETRY — SET 3 · GEOMETRY AND TRIGONOMETRY

Q6

GEOMETRY AND TRIGONOMETRY

A right circular cylinder has a base diameter of 22 centimeters and a height of 6 centimeters. What is the volume, in cubic centimeters, of the cylinder?

- A. 132π
- B. 264π
- C. 726π
- D. $2,904\pi$

Q7

GEOMETRY AND TRIGONOMETRY

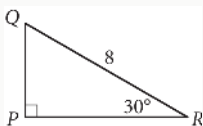
A circle in the xy -plane has its center at $-4, 5$ and the point $-8, 8$ lies on the circle. Which equation represents this circle?

- A. $x - 4^2 + y + 5^2 = 5$
- B. $x + 4^2 + y - 5^2 = 5$
- C. $x - 4^2 + y + 5^2 = 25$
- D. $x + 4^2 + y - 5^2 = 25$

Q8

GRID-IN

GEOMETRY AND TRIGONOMETRY



In the right triangle shown above, what is the length of $\overline{PQ}$?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GEOMETRY AND TRIGONOMETRY

Triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. In triangle ABC , the length of $\overline{AB}$ is 170 and the length of $\overline{BC}$ is 850. In triangle XYZ , the length of $\overline{YZ}$ is 60. What is the length of $\overline{XY}$?

- A. 204
- B. 182
- C. 60
- D. 12

Q10

GEOMETRY AND TRIGONOMETRY

A cube has an edge length of 41 inches. What is the volume, in cubic inches, of the cube?

- A. 164
- B. 1,681
- C. 10,086
- D. 68,921